Title

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Month Day, Year

§1 Preliminaries

§1.1 Definitions

- *Order of a DE:* The highest-order derivative appearing in the DE
- *Linear DE:* a DE in which there are no powers 2 of either the unknown function or its derivatives, and there are no products of the function or its derivatives with each other
- *Initial Value Problem (IVP):* a differential equation that comes with one or more initial conditions for the unknown function and/or its derivatives
- *Interval of Validity:* the largest possible interval on which a solution is valid and meets initial conditions
- *General solution:* a solution which does not take into account initial conditions
- *Actual solution:* a solution which takes into account initial conditions
- *Explicit solution:* a solution which solves for the function in the form $y = y(x)$
- *Implicit solution:* a solution which is not explicit

§1.2 Direction Fields

What can you do with direction fields?

1. Sketch solutions
2. Determine long-term behavior
3. Graph a family of solution/integral curves by starting with different initial values

§1.3 Fundamental questions in Differential Equations

1. Does a solution exist?
2. If a solution exists, is there a unique solution? What information do we need to narrow it down?
3. If a solution exists, can we find it?

§2 First Order Differential Equations

There *is* a formula for solutions, but it is often easier to use the process:

1. Put the equation into this form: $y' + p(t)y = g(t)$ where $p(t)$ and $g(t)$ are both continuous.
2. Find a function $\mu(t)$ called an *integrating factor* which satisfies $\mu(t)p(t) = \mu'(t)$.
   (a) Note that $p(t) = \frac{\mu'(t)}{\mu(t)} = (\ln(\mu(t)))'$.
   (b) Then $\mu(t) = e^{\int p(t)\,dt + k}$
   (c) Simplify by renaming e^k to k: $\mu(t) = ke^{\int p(t)\,dt}$.
3. Multiply the DE by $\mu(t)$ to get $y'(t)\mu(t) + p(t)\mu(t)y(t) = g(t)\mu(t)$.
4. Substitute $\mu'(t)$ to get $y'(t)\mu(t) + \mu'(t)y(t) = g(t)\mu(t)$.
5. Use the product rule to get $(y(t)\mu(t))' = g(t)\mu(t)$.
6. Integrate to get $y(t)\mu(t) + c = \int \mu(t)g(t)\,dt$.
7. Algebra (and changing the sign of c) gives us the solution:

$$y(t) = \frac{\int \mu(t)g(t)dt + c}{\mu(t)}$$